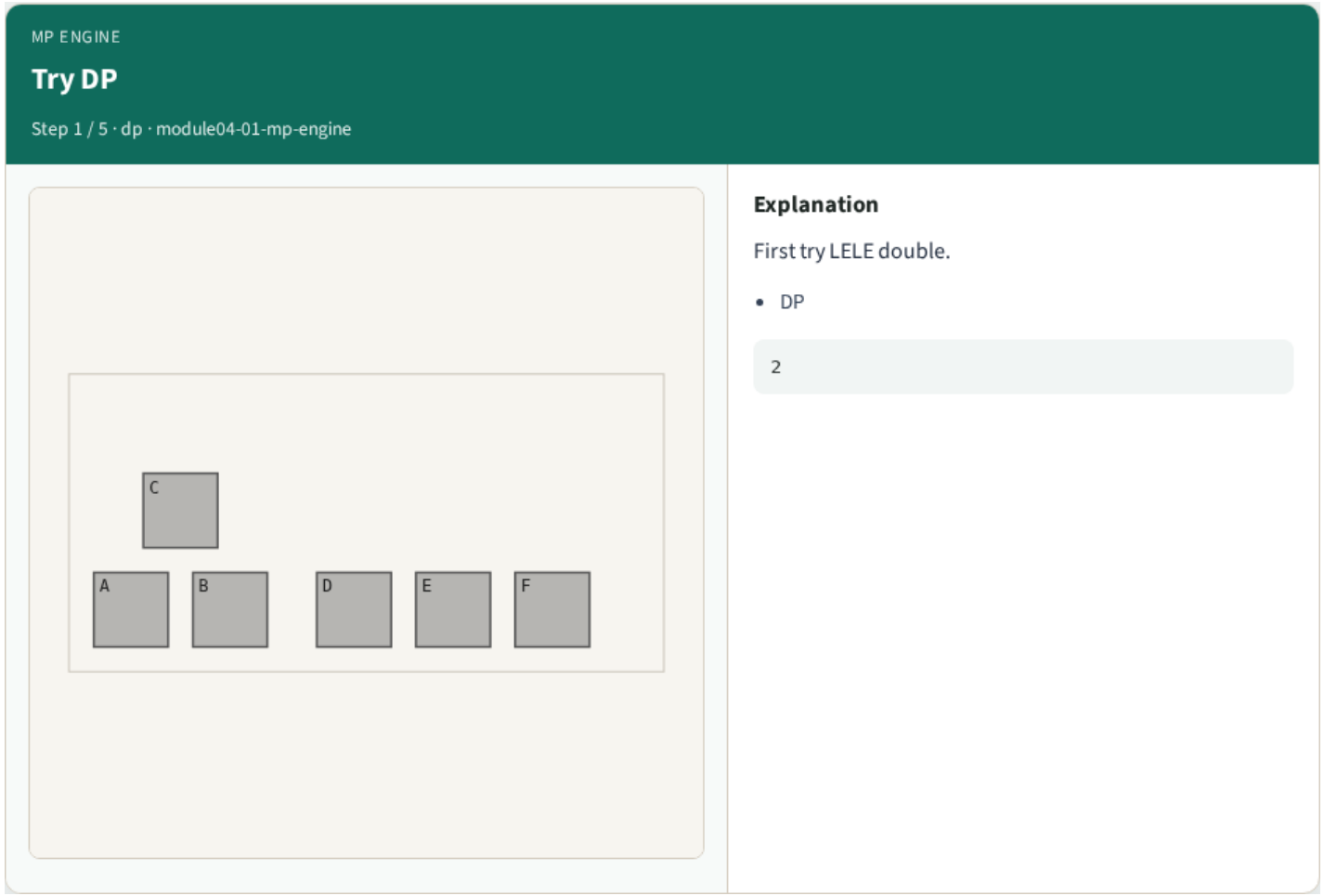
MP engine

A tiny MP engine picks the minimum mask count that succeeds on the layout

The idea

- GOLDEN: best_masks=3, dp_ok=False, tp_ok=True

Try DP


MP ENGINE
Try DP
Step 1 / 5 · dp · module04-01-mp-engine
C
A
B
D
E
F
Explanation
First try LELE double.
DP
2

Try DP

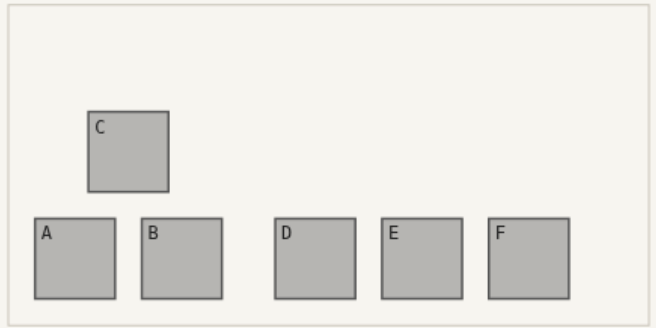
Fail

MP ENGINE

Fail

Step 2 / 5 · fail · module04-01-mp-engine

dp=false



Explanation

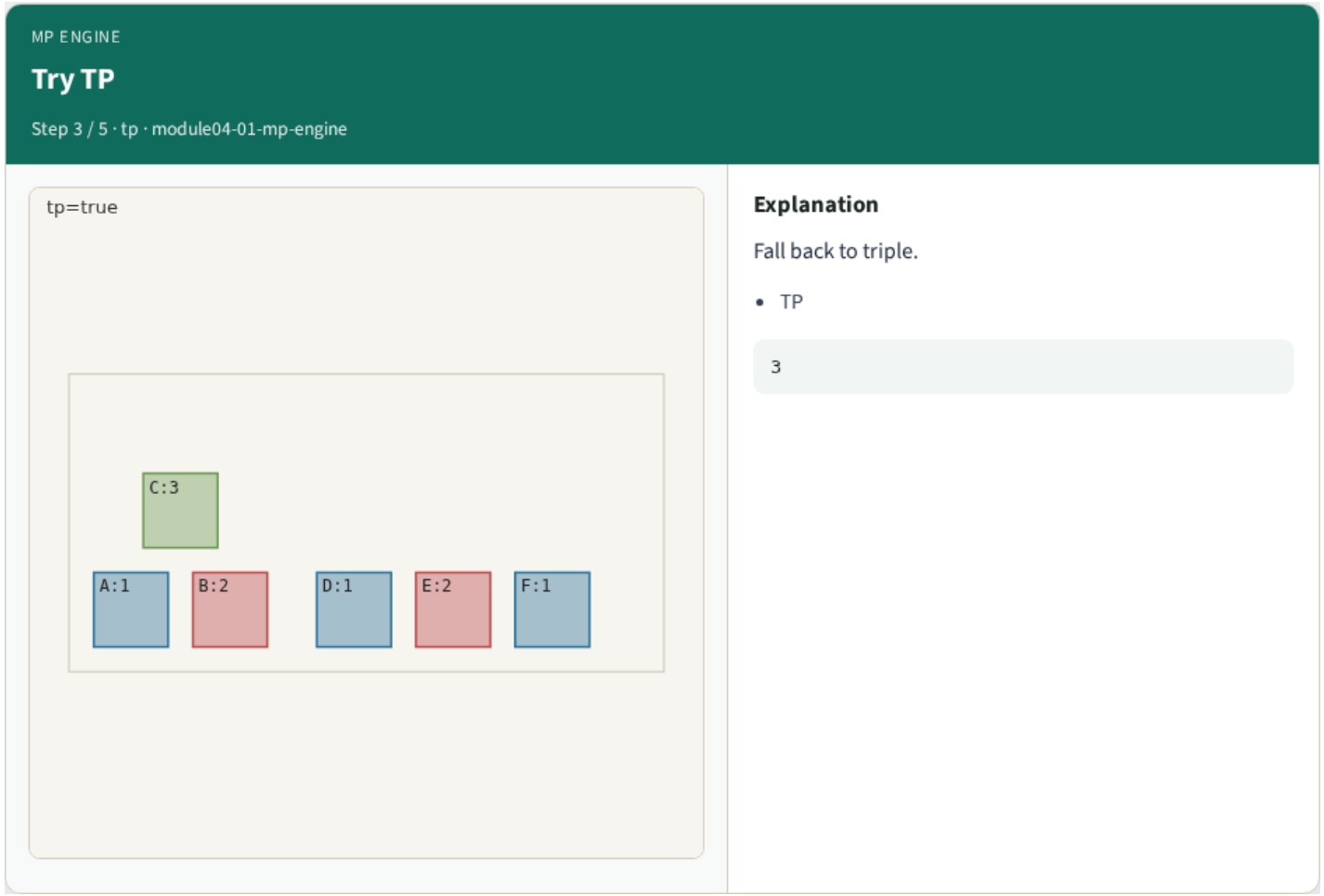
DP fails on the triangle.

- fail

```
ok=false
```

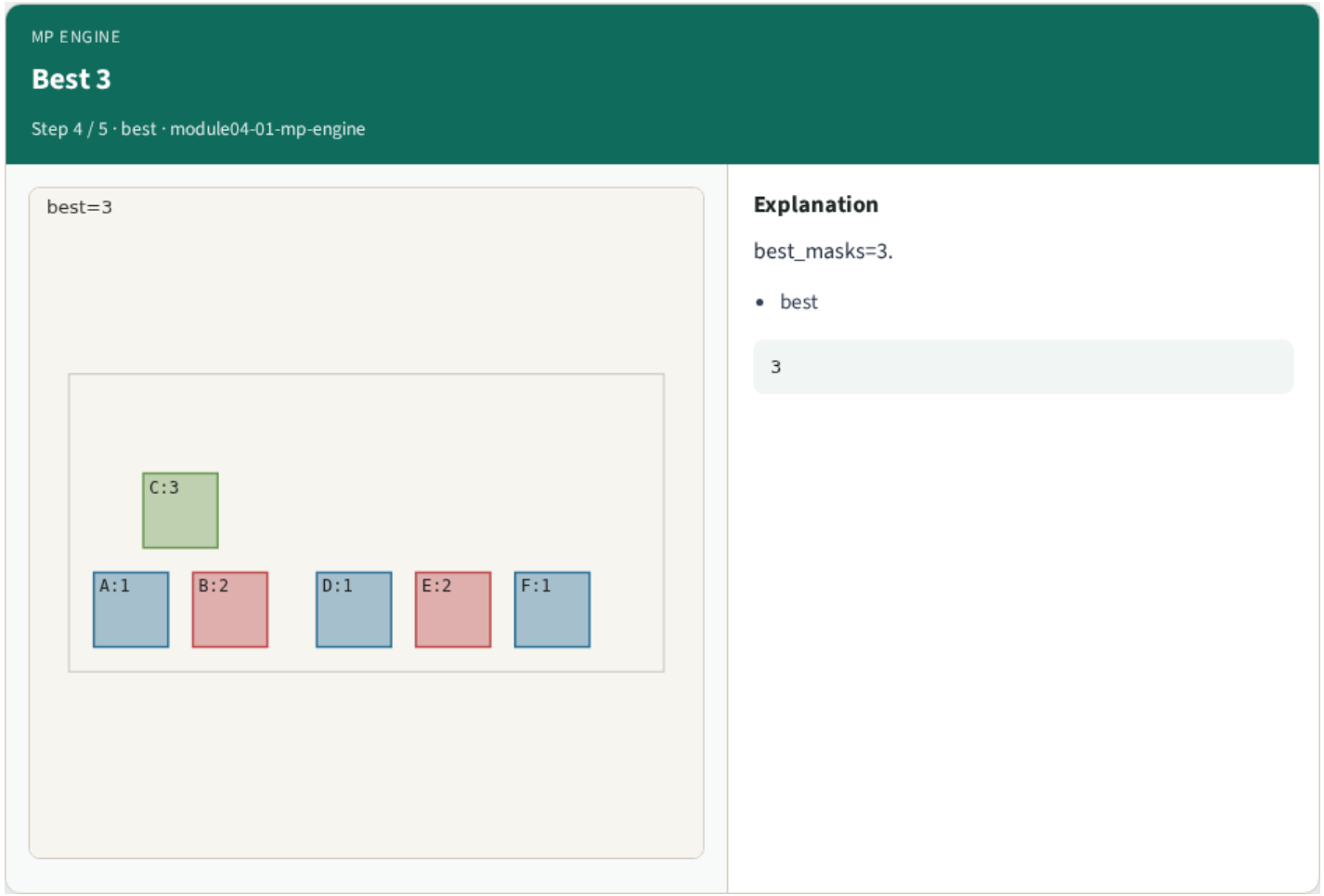
Fail

Try TP


MP ENGINE
Try TP
Step 3 / 5 · tp · module04-01-mp-engine
tp=true
C:3
A:1
B:2
D:1
E:2
F:1
Explanation
Fall back to triple.
TP
3

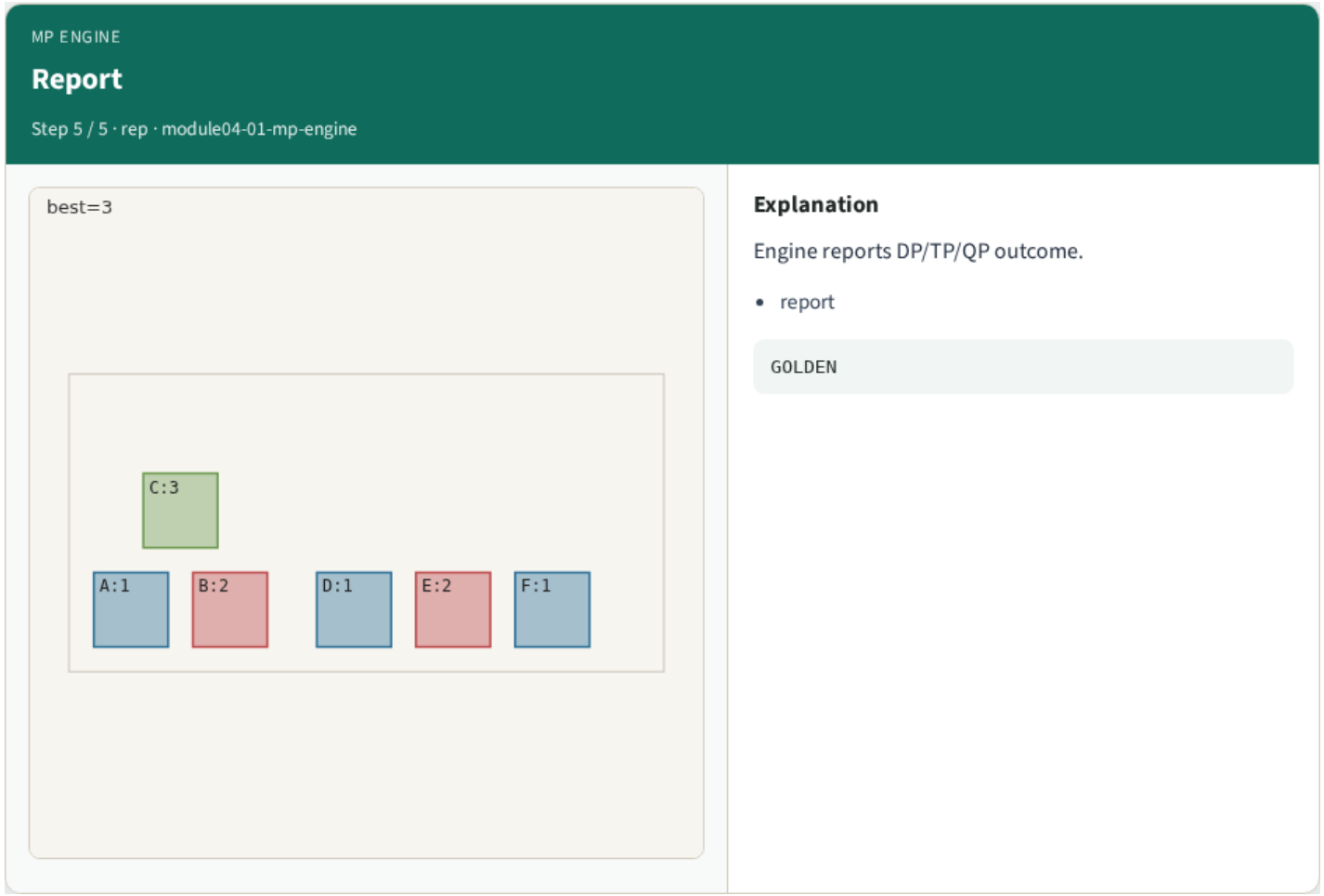
Try TP

Best 3


MP ENGINE
Best 3
Step 4 / 5 · best · module04-01-mp-engine
best=3
C:3
A:1
B:2
D:1
E:2
F:1
Explanation
best_masks=3.
best
3

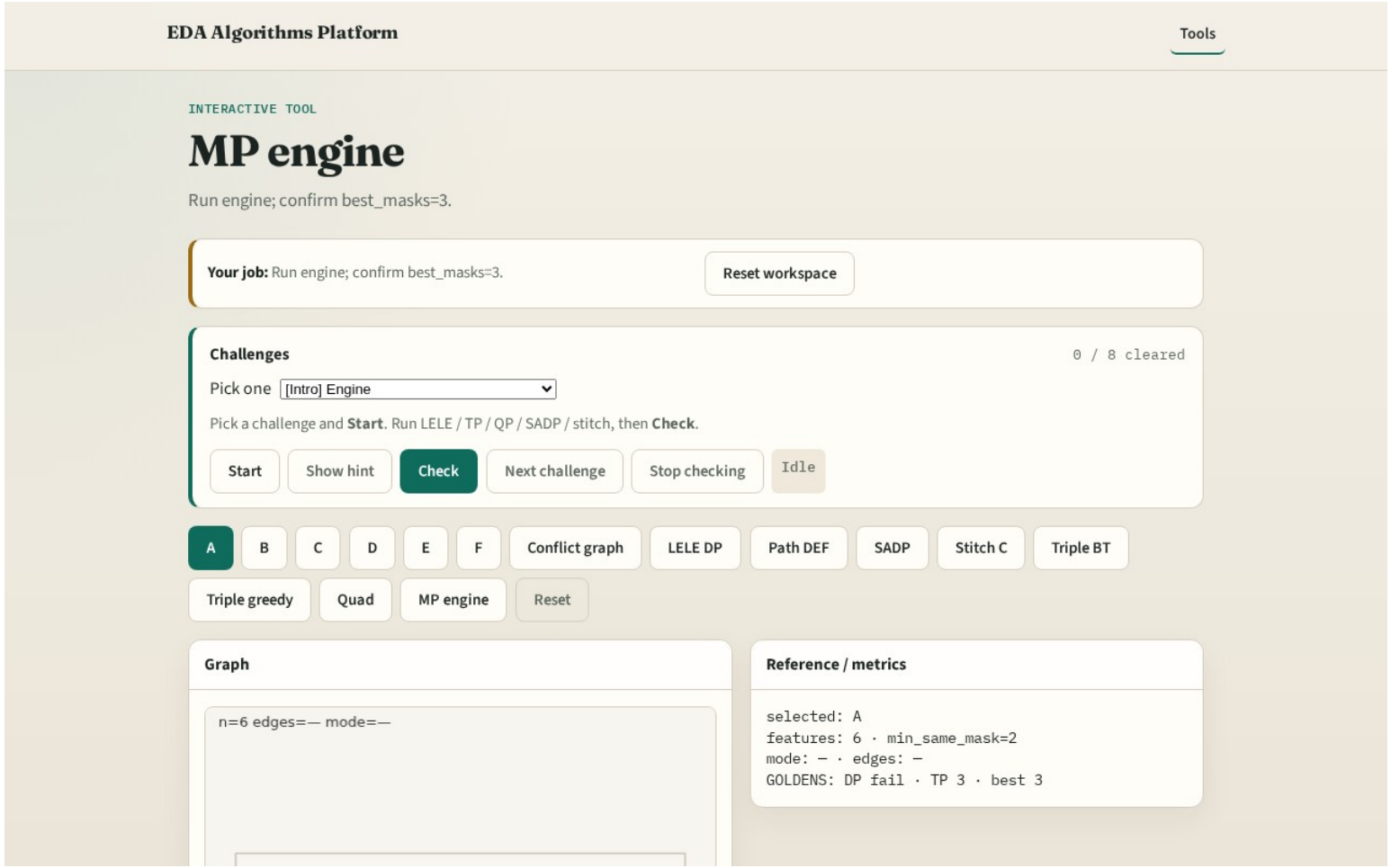
Best 3

Report



Report

Browser lab track



Browser lab starter

Implement track

- Call mp_engine(); assert best_masks==3 and dp_ok is False and tp_ok is True

Pitfalls

- Stopping after DP failure
- Reporting QP when TP already succeeds

Your turn

- Next: offline benchmark compare, then wrap